Canadian Macroeconomic Scenarios: Oil Prices and US Growth, 2026Q3–2028Q3

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Abstract

This note builds three scenarios for the Canadian economy over 2026Q3–2028Q3 and maps them into paths for real GDP, unemployment, CPI inflation, the Bank of Canada policy rate and new house prices. The scenarios differ in two external drivers that Canada does not control: the WTI oil price and US real GDP growth. A vector autoregression (VAR) with these drivers as exogenous variables, estimated on 2009Q3–2026Q2 data, translates each driver path into Canadian outcomes. In the downside scenario, a US slowdown with oil falling to $50 calibrated to the 2001 US recession, Canadian GDP contracts for three consecutive quarters, unemployment peaks near 7.9%, and the policy rate falls close to its effective lower bound by mid-2028. All code and data are available at `https://github.com/Angelgonzales7067/Canada_macro_scenario`.

1 Methodology

1.1 Approach

The exercise follows the workflow commonly used for bank scenario analysis and stress testing: (1) write a short narrative for each scenario; (2) translate it into paths for a few external drivers; (3) use a model estimated on history to map the drivers into domestic variables; and (4) apply judgment where the model is a poor guide.

1.2 Data

The sample is quarterly, 2009Q3 to 2026Q2 (68 quarters). It starts with the first full quarter of the Bank of Canada's daily policy-rate series (V39079, available from April 21, 2009) and ends with the latest quarter for which Canadian and US GDP are published. Monthly series are averaged to quarters before transformation. Growth rates are computed as annualized log differences, $400\,\Delta \ln x_t$; the oil price enters as the quarterly log change, $100\,\Delta \ln P_t$.

1.3 Model

The model is a VAR with exogenous variables (VARX):

$$y_t = c + A_1\, y_{t-1} + B_0\, x_t + B_1\, x_{t-1} + \Gamma\, d_t + u_t, \tag{1}$$

Table 1: Variables, transformations and sources

Variable	Role	Form in model	Source
WTI oil price	Exogenous driver	$100\,\Delta \ln$, quarterly	FRED, DCOILWTICO
US real GDP	Exogenous driver	$400\,\Delta \ln$, annualized	FRED, GDPC1
Canada real GDP	Endogenous	$400\,\Delta \ln$, annualized	StatCan 36-10-0104-01
Unemployment rate	Endogenous	Level, %	StatCan 14-10-0287-01
CPI, all items (SA)	Endogenous	$400\,\Delta \ln$, annualized	StatCan 18-10-0006-01
Policy rate	Endogenous	Level, %	Bank of Canada, V39079
New Housing Price Index	Endogenous	$400\,\Delta \ln$, annualized	StatCan 18-10-0205-01

where y_t contains the five Canadian variables, x_t contains oil-price growth and US GDP growth, and d_t contains dummy variables for 2020Q2 and 2020Q3 that absorb the COVID-19 collapse and rebound. Treating oil and US GDP as exogenous reflects the small-open-economy assumption: Canadian conditions do not move world oil prices or US output within the quarter. It also makes scenario analysis direct, because a scenario is simply a chosen path for x_t.

The lag length is one quarter due to lack of observations. The model is estimated by OLS equation by equation using the R package `vars`.

1.4 Scenario forecasts and judgment

For each scenario, the oil and US GDP paths are fed into the estimated model and the Canadian variables are forecast nine quarters ahead, conditional on those paths. The 2026Q3 oil price is set to its actual average ($87.12) in every scenario. For every scenario, the policy rate was imposed to be greater than or equal to 0.25%

2 Scenarios and historical calibration

Each scenario's driver paths are sized against historical episodes in the same data (Table 3), so that magnitudes reflect what has actually happened rather than arbitrary choices.

Table 2: Scenario assumptions, 2026Q3–2028Q3

Scenario	WTI path (US$/bbl)	US real GDP	Historical calibration
Baseline	Eases from $87 to $75 by 2027Q4, then flat	2.0%	US growth near potential and just below its 2010–19 average (2.4%
Upside	Drifts gently to $70 by 2028Q3	3.0%	2017–19 expansion: US growth averaged about 2.8%, WTI traded between $48 and $70
Downside	Falls to $68 in 2026Q4 and $50 by 2027Q1, then flat (−43%)	0.5%	2001 US recession: US growth averaged 1.1% over 2000Q3–2002Q4, WTI fell 36%; the oil decline is similar to 1998 (−44%)

The downside is a plausible, moderate scenario rather than a severe stress. A severe scenario for stress-testing purposes would be calibrated to 2008–09: oil down roughly 60% and a US contraction.

Table 3: Historical episodes used for calibration (quarterly averages)

Episode	WTI peak to trough	US growth (avg.)	Canada GDP level (peak to trough)	Canada unemp. (change)
1998 Asia crisis	\$23 → \$13 (−44%)	4.4%	No decline	No increase
2001 US recession	\$32 → \$21 (−36%)	1.1%	No decline	+1.0 pt
2008–09 GFC	\$124 → \$43 (−65%)	−0.3%	−4.4%	+2.5 pts
2014–16 oil bust	\$103 → \$33 (−68%)	2.7%	−0.5%	Flat (national)

Source: author's calculations from FRED and Statistics Canada data in the repository.

3 Results

Table 4 reports 2027 averages and the gap from the baseline; Figure 1 shows the full quarterly paths.

Table 4: 2027 averages by scenario (growth rates annualized, %)

	WTI (\$)	US GDP	Canada GDP	Unemp.	CPI	Policy rate	House prices
Baseline	78.4	2.0	1.44	7.11	1.29	1.80	0.12
Upside	79.6	3.0	1.80	6.96	1.29	1.96	−0.05
Downside	50.0	0.5	0.15	7.58	0.42	1.39	−0.74
Upside − baseline	1.3	1.0	0.37	−0.15	0.00	0.16	−0.17
Downside − baseline	−28.4	−1.5	−1.29	0.47	−0.87	−0.41	−0.86

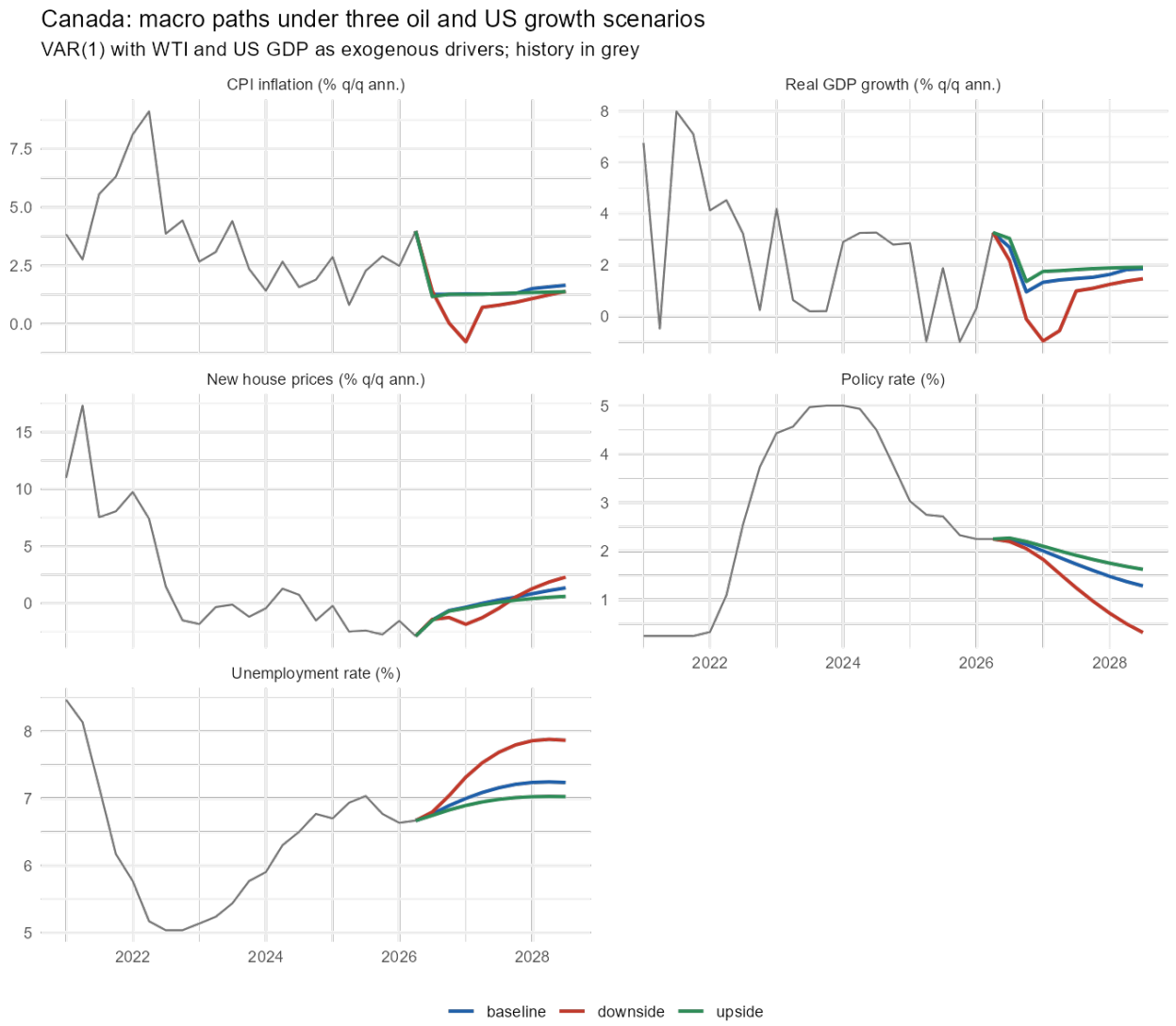


Figure 1: Canadian macro paths under the three scenarios. History since 2021 in grey.

Downside. Canadian real GDP contracts for three consecutive quarters (2026Q4–2027Q2), with a trough of −1.0% annualized in 2027Q1, and the level falls about 0.4% from its peak. This is a mild technical recession, similar in severity to Canada's experience in 2001. Unemployment rises steadily to a peak of about 7.9% in 2028Q2, 1.2 points above its 2026Q2 level of 6.7%. Inflation dips below zero in 2027Q1 as lower oil prices pass through, and the policy rate is cut to about 0.3% by 2028Q3, close to its effective lower bound. New house prices decline through 2027.

Upside. Stronger US demand lifts Canadian GDP growth by about 0.4 points relative to the baseline in 2027, and unemployment is slightly lower. Because the upside oil path declines more slowly than the baseline path, average 2027 oil prices are similar in both, so the upside works almost entirely through the US demand channel.

Supply versus demand shocks. During development, an alternative downside with oil rising to $110 (a supply shock) and US growth of 1.5% was also run. It *raised* Canadian GDP growth relative to the baseline (by about 0.4 points in 2027) and pushed CPI inflation up by about 0.7

points, because Canada is a net oil exporter and higher prices improve its terms of trade. For Canada, the more damaging combination is a demand shock in which oil prices fall *and* US growth weakens, which is why the final downside takes that form. History supports this: in 2014–16 oil collapsed while the US kept growing, and Canadian unemployment did not rise nationally, whereas in 2001 and 2008–09 US weakness was accompanied by rising Canadian unemployment.

4 Limitations

- **Linearity.** The VAR captures average historical relationships. It likely understates responses to large shocks; a 2008-sized shock would probably produce a milder outcome than the -4.4% GDP decline Canada actually experienced.
- **Short sample and imprecise estimates.** The policy-rate series limits the sample to 68 quarters. In the GDP equation, only contemporaneous US growth is statistically significant at the 5% level; oil coefficients are positive but imprecise.
- **Mean reversion.** Forecasts drift toward sample averages. Part of the rise in unemployment in every scenario, including the baseline, reflects reversion toward the 2009–2026 average of about 7%.
- **Missing channels.** The exchange rate, Canadian fiscal policy, and feedback from Canada to the world are not modelled. Forecast uncertainty bands are not reported.
- **Judgment.** The policy-rate floor is imposed by hand, and scenario probabilities are not assigned.